

DIPLOMATIC AND POLITICAL PROBLEMS AFFECTING THE FORMULATION AND IMPLEMENTATION OF AN INTERNATIONAL PROTOCOL FOR ACTIVITIES FOLLOWING THE DETECTION OF A SIGNAL FROM EXTRATERRESTRIAL INTELLIGENCE

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Abstract—This paper is based on interviews with scientists and national space policy makers on what such a protocol should contain and their reactions to the draft of one proposed by the author at the 15th Review Meeting on the Search for Extraterrestrial Intelligence, held during the XXXVII International Astronautical Congress (1986). The key arguments against formulating a post-detection protocol are whether such advance work is necessary or practical, the implications it would have for national sovereignty and the wisdom of disseminating information about SETI research in general to the public. While such problems could be major obstacles to implementing a post-detection protocol, they are not sufficient grounds to discourage efforts to develop one. The principal operational problem will be to find an appropriate and effective international consultative body to which the discovery of extraterrestrial intelligence signals and plans for analyzing and responding to them can be referred. The paper recommends establishment of a new consultative organization which would be chartered under the auspices of the International Astronautical Union or other appropriate international scientific body.

1. INTRODUCTION

The human species is now able to communicate with other civilizations in space, if such exist. Using current radioastronomical technology, it is possible for us to receive signals from civilizations no more advanced than we are over a distance of at least many thousands of light years. The cost of a systematic international research effort, using existing radio telescopes, is as low as a few million dollars per year for one or two decades. The program would be more than a million times more thorough than all previous searches, by all nations, but together. The results - whether positive or negative - would have profound implications for our view of our universe and ourselves.

Thus begins a statement signed in 1982 by 73 prominent scientists from 14 countries urging "...the organization of a coordinated, worldwide, and systematic search for extraterrestrial intelligence[1]." The statement is remarkable because only a few years earlier the international scientific community regarded any "SETI" (search for extraterrestrial intelligence) research program as an object of dubious value; in 1982 the NASA SETI program was abruptly terminated after Senator William Proxmire gave it one of his infamous "Golden Fleece" awards (for US governmental waste and inefficiency). A year later, Senator Proxmire and many critics had done a *volte face*, the NASA program was restored, and by the end of this decade NASA

and JPL will have the capability to scan the entire sky for non-natural radio signals and closely examine some 800 nearby stars similar to our own Sun. At the same time, the technology developed for the search process will accomplish more in 60 seconds "than has been carried out in the past 25 years with all previous programs," according to a recent statement by a NASA specialist[2].

Despite these dramatic changes in attitude, technology and budgets - not to mention the recent intriguing hints from the observation of such stars as Beta Pictoris and Epsilon Eridani that planets may actually exist in other solar systems and could, in fact, be abundant - relatively little has been done to anticipate the political and diplomatic implications of a SETI success[3]. Existing international treaties and agreements on outer space, in particular, do not really address this aspect of space exploration or provide clear precedents for guiding government behavior that would justify leaving the contemplation of what should be done if a SETI search proved successful - or some other evidence of extraterrestrial intelligence were found - to a time when the event occurred[4].

Development of an international protocol for activities following the detection of signal from extraterrestrial intelligence is, thus, a timely and appropriate step for the SETI community to discuss now.

2. FRAMEWORK OF THIS PAPER

This paper discusses certain diplomatic and political problems that could affect the

formulation and implementation of such an international protocol. It is based on interviews with scientists and national space policy makers on what a protocol should contain and their reactions to the draft of one covering the contingency of finding evidence of extraterrestrial intelligence by radio signal as well as by other forms of encounter, proposed by the author at the 15th Review Meeting on the Search for Extraterrestrial Intelligence, held during the XXXVII International Astronautical Federation Congress (1986).

A revised protocol is published as an annex to this paper. The central features of this proposed protocol are the call for the free and open dissemination of information about any such signal and the recommendation of establishing an international consultative process to facilitate its analysis and, if appropriate, formulate a response.

There presently exists a wide variety of attitudes about whether and under what conditions information about a radio contact with extraterrestrial intelligence should be revealed. Attitudes among scientists range from a belief that even preliminary or ambiguous results should be shared widely, to those who think the release of information should depend on the context of the signal or message, the certainty of their findings, and the extent to which the information would adversely affect public order[5]. And despite the 1971 declaration by US and USSR scientists that "the search for extraterrestrial intelligence should be made by representatives of the whole of mankind[6]" some messages have been sent into space by US scientists without international consultation and even by some entrepreneurs who offer to beam up 25 words of a message to the stars into space for \$9.98.

Such attitudes and developments - coupled to finding virtually no appropriate international treaties and agreements that can readily be applied to SETI - led me to propose an international protocol for handling contacts and communication with extraterrestrial intelligence at the Innsbruck symposium and to pursue development of it with US and other governmental authorities and scientific experts since that meeting.

3. THE NEED FOR A PROTOCOL

While the reaction to the specific tenets of such a protocol has been positive, some

scientists as well as diplomats have questioned the need for a code of conduct at all and the practicality of formulating one in advance of a contact[7]. The arguments against developing a code now are that the conditions surrounding a success in SETI cannot really be anticipated and that, in any case, there will be plenty of time for nations to consult in the event of success since any response would probably require years to reach its destination. But I see the function of international law as precisely one of addressing, in part, situations before they arise, especially if some ambiguity or confusing circumstances can be expected. Indeed, much of present-day outer space law addresses situations that have not yet occurred. For example, two UN outer space treaties deal with the type of assistance to be given astronauts in the event of an emergency landing on the territory of a state other than the one from which they were launched[8]. The 1967 treaty also requires the signatories to provide "information on any phenomenon discovered in outer space" which could constitute a danger to the life or health of astronauts (Article V).

I also think that development of an international protocol for SETI is timely and appropriate given not only projected advances in the technology applied to the search described above, which increase the possibilities of success, but also the increase in the number of countries where research is being conducted[9] and the unfortunate tendency by some of these countries and their allies to politicize many outer space issues in existing international forums such as the UN[10]. Pressures of the latter sort, moreover, will only increase as the superpowers deploy SDI systems[11].

Establishing a protocol, especially one that develops a mechanism for international consultation, could allow such a process to gain international scientific and diplomatic acceptance. This would enhance its effectiveness and legitimacy in the event of a success in SETI, when it might be much harder for nations - especially rivals - to agree on modalities for international consultation.

4. POSSIBLE OBJECTIONS TO A PROTOCOL

In addition to general reservations about the need for developing and having a protocol established in advance of what, to many, is an unlikely event, there are several possible objections to specific provisions of my draft protocol that should be addressed.

One of the basic principles of the protocol is that nations should refrain from acting in response to a suspected signal from extraterrestrial intelligence (or other comparable discovery) unilaterally. A possible objection to this aspect of the code - one actually raised in the discussion session at the Innsbruck symposium - is that it infringes on national sovereignty by denying states freedom of action in dealing with an event (e.g., the detection of a signal) which happens in its territory of the discovery of extraterrestrial artifacts on another planet by astronauts or probes manned and operated solely by one government. In an absolute sense, the protocol does infringe on sovereignty but no more than many other facets of international law which nations have accepted, especially in environmental and other outer space matters[12].

In the 1967 UN outer space treaty, moreover, the principle of non-appropriation of outer space (Article II) is "the antithesis of the principle of absolute and exclusive sovereignty" and yet is the most widely accepted of all the UN space treaties and serves as "one of the pillars of international space law[13]." And, in general, sovereignty has come to be accepted by most states as having limits, especially when applied to a number of global issues.

The proposal does involve a measure of regulation of non-government scientific activities and may, in this sense, be seen as an infringement on the sovereignty and freedom of action of the international scientist. Such regulation would be no more extensive or intrusive than that which presently exists or that which countries already accepted when they agreed in the 1967 UN Treaty to "bear international responsibility for national activities in outer space... whether such activities are carried on by governmental agencies or non-governmental entities (Article VI)." In any case, the requirement of the protocol for international consultation in the event of a detection of an extraterrestrial radio signal should be acceptable if - as I propose - it is designed and is adopted first by the international scientific community and then by governments and international organizations.

Another possible objectionable feature of the code is the requirement for the free and open dissemination of information about the detection of a signal (or some other evidence) of extraterrestrial intelligence. The arguments against free and open dissemination stem largely from three sources. One is a concern that the

discovery that we are not alone could cause panic in some societies or trigger a "War of the Worlds" type reaction. While a potential for this does exist, the experience of the past 30 years in announcing space discoveries is certainly a precedent for thinking that the public would be able to cope with the knowledge that another intelligent civilization or life form exists in the universe. Highly sophisticated bodies of philosophical and religious thinking in both Eastern and Western societies have anticipated the discovery of extraterrestrial intelligence for nearly 2500 years[14]. Public opinion polls in several industrialized countries suggest that there is already widespread belief in the existence of extraterrestrial intelligence, and religious and social systems in many primitive societies also apparently include acceptance of the belief that life on earth has been affected by forces beyond those visible in the immediate environment.

A second concern about the requirements for free and open dissemination of information about a successful SETI contact stems from the tendency of some scientists to want to make absolutely certain that their findings are correct - or at least beyond ridicule. The discovery of pulsars, for example, was not announced immediately, in part because British scientists wished to be certain of their findings. But the mysteries of the data would have been unravelled sooner had the findings been widely disseminated. And since, I suspect, there is a high probability that the first SETI results will be ambiguous or confusing, there is all the more reason to enlist the largest number of experts possible in interpreting what has been found.

Free and open dissemination of a SETI discovery, finally, could be inhibited if the signal were detected through a classified military or intelligence communication or research facility. In this event, there is very little that an international protocol could do to require governments to surrender vital national security secrets. But, I think, the pressures not to keep a SETI success secret could well outweigh security concerns and most government leaders would be disposed to find a way to make the information public, especially if there were - via the protocol - a standing international precedent and mandate for doing so.

The usefulness and effectiveness of a post-detection international protocol will necessarily be subject to the vagaries of international politics and especially the level of tension in the international system. However, periods in which global or regional tensions are high are striking

for the degree to which most international agreements are honored rather than violated. Tensions in the US-Soviet relationship have been very high for the better part of the past ten years. Yet the signed, but not ratified, Salt II agreement has proved remarkably durable and, until recently, has been characterized by a very high degree of compliance. To take another example, despite the extremely poor US-Iran relationship, the two countries have accepted international law and the arbitral process established to resolve their claims against each other.

The key thing to keep in mind is that international agreements are honored as long as they serve national interests, especially in periods of great tension. Most states have a vital interest in avoiding warfare and in containing the impact of the situation which gave rise to the tension or crises in the first place. I think the same dynamics would apply to an international protocol on SETI; even in times of considerable international discord, the sharing of information about SETI should prove as important to national interests as the sharing of information about earthquakes or disease epidemics.

While such problems as those highlighted above could be obstacles to implementing a post-detection protocol, they are not sufficient grounds to discourage or abandon efforts to develop one. From a political viewpoint, success in the detection of an extraterrestrial intelligence signal is likely to transcend national rivalries as has generally been the case in space exploration. Formulation of an international protocol now can help to codify further a pattern of governmental cooperation in space activities and the tendency toward the free and open dissemination of information about scientific activities and discoveries. Moreover, adherence to a widely-accepted and long-established protocol is likely to be high, regardless of the level of tension in the international system or the process by which a signal of extraterrestrial intelligence is detected.

5. A MECHANISM FOR IMPLEMENTING A PROTOCOL

If a post-detection protocol is developed along the lines suggested above, it should also include an appropriate international consultative body to which the discovery of extraterrestrial intelligence signals and plans for analyzing and responding to them can be referred.

Unfortunately, the problems associated with most international organizations today are formidable barriers to using them. While the United Nations is a logical place to turn, it has become deeply politicized over both internal administrative matters and many international issues.

An alternative to using an existing politicized international organization would be the creation of a Standing consultative Commission on SETI. This commission could be chartered under the auspices of the International Astronautical Union or another appropriate body and would meet at least annually. Extraordinary sessions of the commission could be convened to consider research findings that appear to indicate the existence of extraterrestrial intelligence. The commission should also build on the important work of Dr. Jill Tarter by requesting an annual report from each on-going SETI research project and publishing a compendium of such reports.

The membership of the commission should both be representative of the governments and private scientific organizations conducting SETI research and interdisciplinary researchers (including, for example, scientists, social scientists, and philosophers), who would serve five year terms, without compensation but with reimbursement for travel to annual and special meetings. A small secretariat would be established to schedule commission meetings and to receive and prepare annual and special reports for publication. These administrative costs should be financed by an endowment raised from donations by governments and private persons and foundations.

6. ANNEX

Proposed International Protocol for Activities Following the Detection of a Signal From Extraterrestrial Intelligence:

Recognizing that the search for extraterrestrial intelligence is an integral part of space exploration and is being undertaken for peaceful purposes and for the common interest of all mankind,

Inspired by the profound significance for mankind of detecting evidence of extraterrestrial intelligence,

Recalling the United Nations treaty on Principles Governing the Activities of States in the Exploration and use of Outer Space, including the Moon and other Celestial Bodies which commits the States parties to this treaty "to inform the Secretary General of the United

Nations as well as the public and the international scientific community, to the greatest extent feasible and practicable, of the nature, conduct, locations, and results" of their space exploration activities (Article XI), the following guidelines for disseminating information about and handling contacts with extraterrestrial intelligence should be observed by all scientists and governments:

1. Any individual, public or private astronomical observatory, or governmental authority which has detected a signal from or other evidence of extraterrestrial intelligence will freely and openly announce their findings and submit all data necessary for independent verification to the international scientific community.
2. Plans for analyzing the evidence of an extraterrestrial signal and any response to an actual communication form an extraterrestrial intelligence will be formulated by a Standing Consultative Commission chartered under the auspices of the International Astronautical Union.
3. In the event that a signal from extraterrestrial intelligence appears to pose a threat to human health or peace, nations will immediately consult the Security Council of the United Nations.

REFERENCES

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2. Peter Backus, quoted in a *Reuters* dispatch, 23 March 1987. There have been some 46 serious radio and optical astronomical SETI searches since 1960.
3. For an analysis of some of the international legal and diplomatic implications of dealing with extraterrestrial intelligence in a sustained way - as opposed to the focus of this paper, which is on how to deal with the very first contact - see the pathbreaking work of Ernst Fasan, *Relations with Alien Intelligences: The Scientific Basis of Metalaw*. Berlin Verlag (1970), and Michael Michaud, especially his Interstellar Negotiations, *Foreign Service Journal* 10, p. 30 (December 1972) and Extraterrestrial Politics. *Cosmic Search*, 28-30 (Summer 1979).
4. A. E. Goodman, The Diplomatic Implications of Discovering Extraterrestrial Intelligence. *Mercury* XVI, 56-57 (March-April 1987).
5. For a discussion of the rationales for these views, see Alan Tough, "A Critical Examination of Factors That Might Encourage Secrecy," paper number IAA-87-586, prepared for the 38th Congress of the International Astronomical Federation (October 1987) (paper contained in this special Issue); Ian Ridpath, *Messages From The Stars*. Harper and Row, New York (1978) and Kendrick Frazier, First Contact: The News Event and The Human Response. *Extraterrestrial Intelligence: The First Encounter*. Prometheus Books, Buffalo, pp. 79-81 (1979).
6. Carl Sagan, ed., *Communication With Extraterrestrial Intelligence*. MIT Press, Cambridge, p. 354 (1973).
7. As professor Stephen Gorove has pointed out, not all scientific and technological advances necessarily lead to international agreements or new space laws. See his Expectations in Space Law: A Peek into the Future. *J. International Affairs*, p. 187 (1986).
8. The Treaty on Principles Governing the Activity of States in the Exploration and Use of Outer Space, including the Moon and other Celestial Bodies - adopted on 19 December 1966, opened for signatures on 27 January 1967, entered into force on 10 October 1967, and the Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space - adopted on 19 December 1967, opened for signatures on 22 April 1968, entered into force on 3 December 1968.
9. At present the US, USSR, West Germany, the Netherlands, Japan and Canada are funding SETI radio telescope searches.
10. For details see the US Department of State's assessment of COPUOS in Aeronautics and Space Report of the President, National Aeronautics and Space Administration, Washington, D.C., pp. 107-108 (1986).
11. S. F. Wells, Jr. and R. S. Luttwak, *Strategic Defenses and Soviet-American Relations*. Ballinger Pub. Co., Cambridge (1987).
12. See the conclusions reached in: Imre Csabafi, *The Concept of State Jurisdiction in International Space Law*. Martinus Nijhoff, The Hague, pp. 152-155 (1971).
13. Ibid, 116-117. See also: C. Q. Christol, *The Modern International Law of Outer Space*. Pergamon Press, New York, pp. 241-361, pp. 841-848 (1982).

14. For a brief review of this literature, see M. D. Papagiannis, A Historical Introduction to the Search for Extraterrestrial Life. *The Search for Extraterrestrial Life: Recent Developments*. D. Reidel Pub. Co., Dordrecht, Holland, pp. 5-10 (1985).